Assignment
CCS23213 Data Mining
Real-World Data Mining Project

No Matrics:

Criteria	Excellent (5)	Very Good (4)	Good (3)	Satisfactory (2)	Weak (1)	Marks
1. Problem Identification and Dataset Selection	Dataset background is clearly explained. Source, number of records, number of attributes, and measurement scale of each attribute are accurately identified. The problem, objective, and data mining task are clearly defined.	Dataset background and structure are well explained with minor weaknesses. Problem, objective, and task are mostly clear.	Dataset is explained adequately, but some details are unclear. Problem and objective are stated but lack depth.	Basic explanation is given, but important dataset details or problem statement are missing.	Dataset description is very limited. Problem, objective, and data mining task are unclear or inappropriate.	
2. Algorithm Selection and Justification	Two suitable algorithms are selected and strongly justified based on the problem and dataset.	Two suitable algorithms are selected with good justification and minor weaknesses.	Algorithms are acceptable, but justification is limited.	Algorithm choice is weak or only partly suitable for the problem.	Algorithm choice is missing, inappropriate, or not justified.	
3. Handling Missing Values	Missing values are correctly detected and handled using suitable methods. Clear justification and results are provided.	Missing values are properly handled with good explanation and minor weaknesses.	Missing values are identified and handled, but explanation or justification is limited.	Missing values are only partly addressed or weakly explained.	Missing values are not properly detected or handled.	
4. Outlier Detection and Handling	Outliers are clearly detected using suitable methods and visualizations. Handling is appropriate and interpretation is accurate.	Outliers are detected and handled well with minor weaknesses.	Outliers are identified and handled at an acceptable level, but interpretation is limited.	Outliers are partly identified, but handling or interpretation is weak.	Outliers are not properly detected, handled, or interpreted.	
5. Data Distribution and Interpretation	Data distribution is clearly examined using suitable statistics and visualizations. Skewness is correctly identified and interpreted meaningfully.	Data distribution is examined well and interpretation is mostly correct.	Data distribution is examined adequately, but interpretation lacks depth.	Basic examination is shown, but interpretation is weak.	Data distribution is not properly examined or interpreted.	
6. Normalization	Appropriate normalization method is applied correctly with strong justification based on the dataset and algorithm used.	Normalization is applied correctly with good explanation and minor weaknesses.	Normalization is generally appropriate, but explanation is limited.	Normalization is applied at a basic level with weak justification.	Normalization is missing, inappropriate, or poorly explained.	
7. Feature Selection	Relevant features are selected appropriately with clear justification based on the analysis method and model requirement.	Features are selected appropriately with minor weaknesses in justification.	Feature selection is acceptable, but explanation is limited.	Feature selection is weak or only partly justified.	Feature selection is missing or inappropriate.	
8. Exploratory Data Analysis (EDA)	EDA is clearly conducted using suitable summary statistics and visualizations such as histograms, scatterplots, box plots, and correlation matrices.	EDA is conducted well with relevant graphs and minor weaknesses in explanation.	EDA is adequate, but some visualizations or explanations are limited.	EDA is basic and lacks important analysis or suitable visualizations.	EDA is missing, inappropriate, or not properly presented.	
9. Interpretation of EDA Findings	EDA findings are interpreted clearly and meaningfully. Patterns, relationships, outliers, and insights are well explained and linked to further analysis.	EDA findings are interpreted well with minor weaknesses.	Interpretation is adequate, but insights are limited.	Some interpretation is provided, but it is weak or unclear.	Interpretation is very limited, inaccurate, or missing.	
10. Model Building and Experimentation	Two algorithms are correctly implemented. Five train-test splits are completed: 50:50, 60:40, 70:30, 80:20, and 90:10. The experiment is systematic and clearly presented.	Model building is done well with minor weaknesses in implementation or presentation.	Model building is acceptable, but some steps or explanations lack clarity.	Model building is basic, incomplete, or contains several weaknesses.	Model building is incorrect, incomplete, or not clearly shown.	

11. Model Evaluation and Comparison	Appropriate metrics are used correctly. Results are clearly compared between algorithms and train-test splits.	Evaluation and comparison are mostly correct with minor weaknesses.	Evaluation is acceptable, but comparison or explanation is limited.	Evaluation is basic and lacks proper comparison or justification.	Evaluation metrics are missing, incorrect, or poorly explained.	
12. Best Model Selection and Validation	Best split for each algorithm and best overall model are clearly identified and strongly justified. The best model is validated using five unseen input examples with predicted output.	Best model selection, and validation are done well with minor weaknesses.	Best model is selected, but justification, or validation is limited.	Best model selection is weak and validation is incomplete or unclear.	Best model selection or unseen input validation is missing or inappropriate.	
13. Findings, Recommendations, Strengths, and Limitations	Findings are summarized clearly. Recommendations are practical and based on results. Strengths and limitations are discussed thoughtfully.	Findings and recommendations are clear with minor weaknesses.	Discussion is adequate, but recommendations or limitations lack depth.	Basic discussion is provided, but it is limited or weakly linked to results.	Discussion, recommendations, strengths, or limitations are missing or unclear.	
14. Report Presentation and Organization	Report is well organized, clear, neat, and professional. It follows the required structure: Introduction, Methods, Results, Discussion, Conclusion, References, and Appendix if necessary.	Report is clear and organized with minor issues.	Report is generally understandable, though some parts lack clarity or organization.	Report shows weak organization, formatting, or writing clarity.	Report is poorly organized and difficult to follow.	
15. Presentation and Understanding	Presentation is clear, well organized, and confidently delivered. Students explain the problem, dataset, methods, results, and conclusions effectively. All members participate, answer questions well, and demonstrate that the work is their own, not solely generated by ChatGPT or other AI tools.	Presentation is clear and well delivered with minor weaknesses. Most members participate, answer questions adequately, and show good understanding of their own work.	Presentation is acceptable, but explanation, delivery, or participation is limited. Students show some understanding but rely too much on slides or prepared text.	Presentation is basic, lacks clarity, or shows weak understanding. Students have difficulty explaining the work, suggesting over-reliance on AI-generated content.	Presentation is poorly delivered, incomplete, or students are unable to explain the project. Work appears to be copied or generated by ChatGPT or other AI tools without proper understanding.	
					Total	**/75**
						/20